



Derick Jose's tips to progress along the maturity curve of metaverse deployment

- Step 1: Map the high value applications of metaverse which are ripe for embedding into business process.
- Step 2: Nail the granular business case and ROI model.
- Step 3: Crawl-walk-run. Start with a 90-day pilot and then scale.

(Source: YourStory)

intelligent troubleshooting. This way, they can record deviations in existing processes and suggest next steps for improvement to their employees. They use TeamViewer Frontline, tailored to the hospitality, retail, and food supply chains.

Kemper GmbH, a technology leader for extraction and filter systems in the metalworking industry, uses TeamViewer Frontline for AR based remote support for direct communication in customer service. Experts communicate live with the customers via smart glasses and thus guide them quickly and easily through the problem-solving process.

Airbus Helicopters Inc. uses TeamViewer Frontline in quality assurance and documentation to improve their operations. With the help of smart glasses equipped with a camera and microphone, important data, for example during the maintenance of gearboxes, is documented easily and quickly. Images and voice recordings are uploaded and saved directly in the SAP system via TeamViewer's solution.

German railway operator Deutsche Bahn's industrial metaverse, developed using solutions from Nvidia and Siemens, will be based on a digital twin that represents 5,700 stations and tracks covering more than 30,000km. The digital twin will rely on a huge network of sensors working on 5G wireless as well as cameras and lidar detection systems fitted on the trains. The data collected by these devices will be aggregated and used to train their AI system, which in turn will help them understand what is happening, iden-

tify problems, and fix them sooner.

In Berlin, Siemens and Bentley Systems are using digital twin technology to create a city district, which will be both 100% CO²-neutral and 100% barrier-free. In London, Thameslink uses Siemens technology to ensure 100% availability of its fleet.

Hellenic, one of Coca Cola's largest bottlers, serves 29 markets across Europe. It has more than 55 facilities across Europe and produces 90,000 bottles of Coca Cola products per hour on one production line alone. The company has implemented industrial metaverse capabilities with Microsoft to optimise their processes and improve efficiency. According to reports, the factory saved over 9% in energy cost in a 12-week period by building out a sensor fabric, creating digital twins, and allowing factory workers to become immersed in the twins.

Siemens has built a digital native plant in Nanjing, China. The plant's performance was simulated with a digital twin at the design stage itself. This helped avert planning errors, much before physical construction began. For example, the planning team was able to spot a painting machine without proper ventilation during a virtual inspection. This would have resulted in a lot of trouble in the real-world, but because they detected it in the digital twin in the design phase itself, all it took was a few minutes to change the plan! According to the company, the digital twin has helped the company increase manufacturing capacity by 200% and productivity by 20%.

Anheuser-Bush InBev (AB InBev), the world's largest brewer with 200 breweries and over 150,000 employees across the globe, is using a private industrial metaverse to transform its operations. They have created a comprehensive digital model of their breweries and supply chain using Azure Digital Twins. The digital model is synchronised with operational and environmental data, to reflect the complex relationships between the natural ingredients and the brewing process. This enables brew masters to adjust settings based on active conditions. Mixed reality overlays help frontline operators to get assistance from remote experts, when required. Deep reinforcement learning helps packing line operators to spot and fix bottlenecks in canning and bottling operations, and schedule maintenance, without much downtime.

Renault Group recently announced the launch of its industrial metaverse, which it describes as a "true replica of the physical world controlled in real time." All of the group's production lines are now connected (8,500 pieces of equipment), 90% of supply flows are constantly monitored, and 100% of supply chain data is hosted in the Renault Group Metaverse. The company claims that by 2025, its metaverse will generate savings of Euro 320 million, plus Euro 260 million in inventory savings, a 60% reduction in vehicle delivery time, 50% reduction in the carbon footprint of vehicle manufacturing, and a contribution to the 60% reduction in warranty costs targeted by the whole group.

Kawasaki's DX initiative uses Microsoft's industrial metaverse technology to implement a real-time, virtual approach to robot care and maintenance. This technology allows Kawasaki to quickly solve robot failures, and also prevent issues using remote predictive main-